

DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

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		DM001-D-GAB-WSC-DR-DCC-545046	A1

DUBAI RAPID LINK CONSORTIUM

**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

**Siding & Cleaning Building
(Building 1)**

**Supplementary Structural Design Calculations
(DCP4 Submission)**

June 2008

Client: JT Metro JV	Contract No. (if any):
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Covering Letter / Transmittal Ref. No.:	Date of Issue: 11 June 2008

Revision, Review and Approval/Authorization Records

A1	DCP4 Submission	D. Shah June 08	M. Lake June 08	J. Hill June 08
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

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1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (Paul Shepherd-Smith)
4	Atkins (Dipak Shah)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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1 Introduction

The design report is submitted as a supplementary calculation for capacity , connection design of Curve Purlin IPEA 120 & Buffer Stop Design.

Please refer to previously submitted Document no. DM001-D-GAB-WSC-DR-DCC-544954 Rev A2 for the design calculations of all other structural members, which are not covered in this document.

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2. Curve Purlin Design

DISPLACEMENTS SUMMARY

	Node	L/C	Horizontal	Vertical	Horizontal	Resultant	Rotational		
			X mm	Y mm	Z mm	mm	rX deg	rY deg	rZ deg
Max X	3	301 SELF+DL	2.541	-0.335	0	2.563	0	0	0.037
Min X	5	304 0.8(LL+WL2)	-0.075	-1.612	0	1.614	0	0	0.027
Max Y	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Min Y	6	301 SELF+DL	-0.067	-2.21	0	2.211	0	0	-0.071
Max Z	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Min Z	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Max rX	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Min rX	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Max rY	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Min rY	1	301 SELF+DL	0	0	0	0	0	0	-0.137
Max rZ	4	301 SELF+DL	1.147	-0.88	0	1.446	0	0	0.118
Min rZ	7	301 SELF+DL	0	0	0	0	0	0	-0.157
Max Rst	3	301 SELF+DL	2.541	-0.335	0	2.563	0	0	0.037

MAXIMUM DEFLECTION

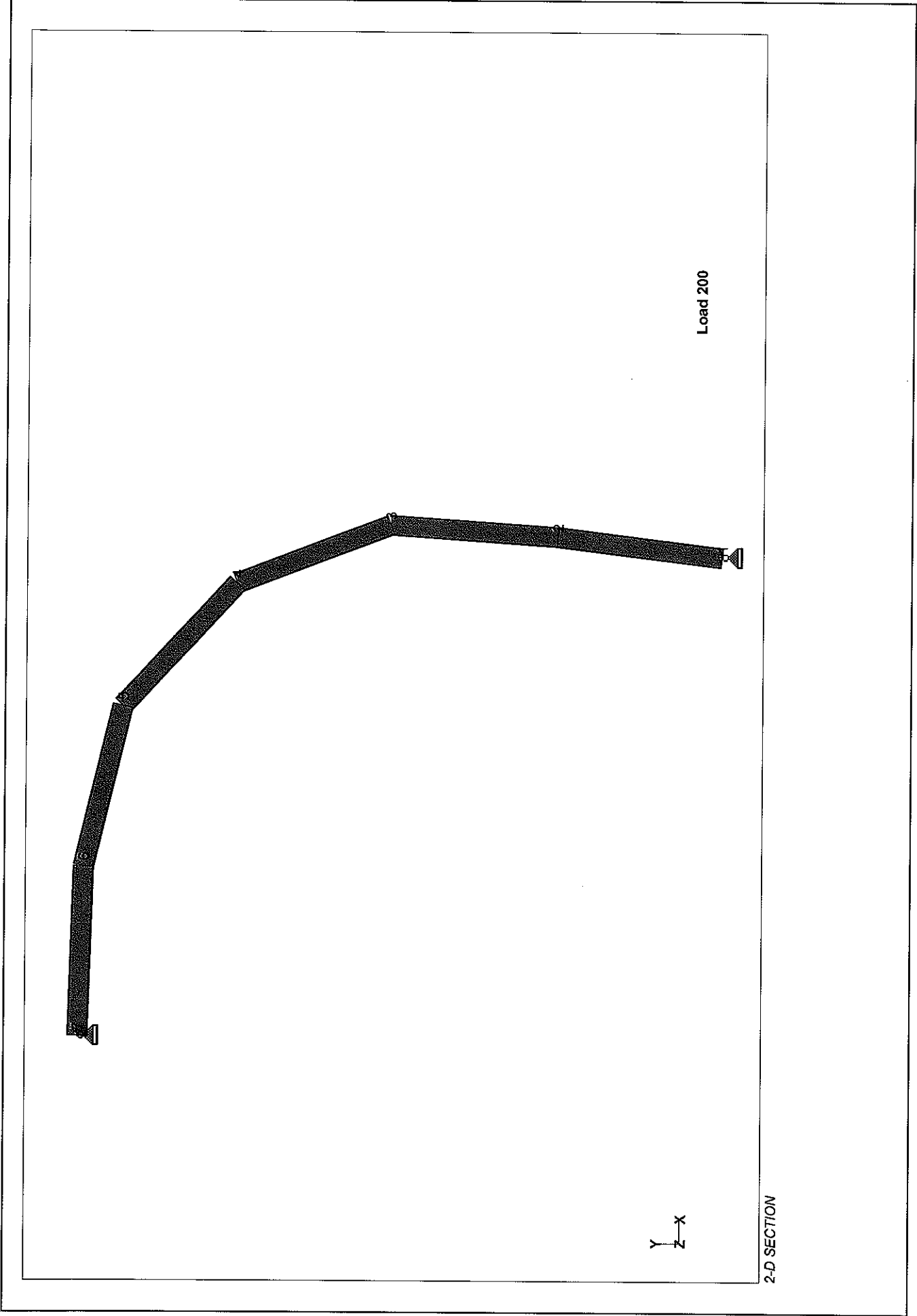
ALLOWABLE DEFLECTION (IN MM)

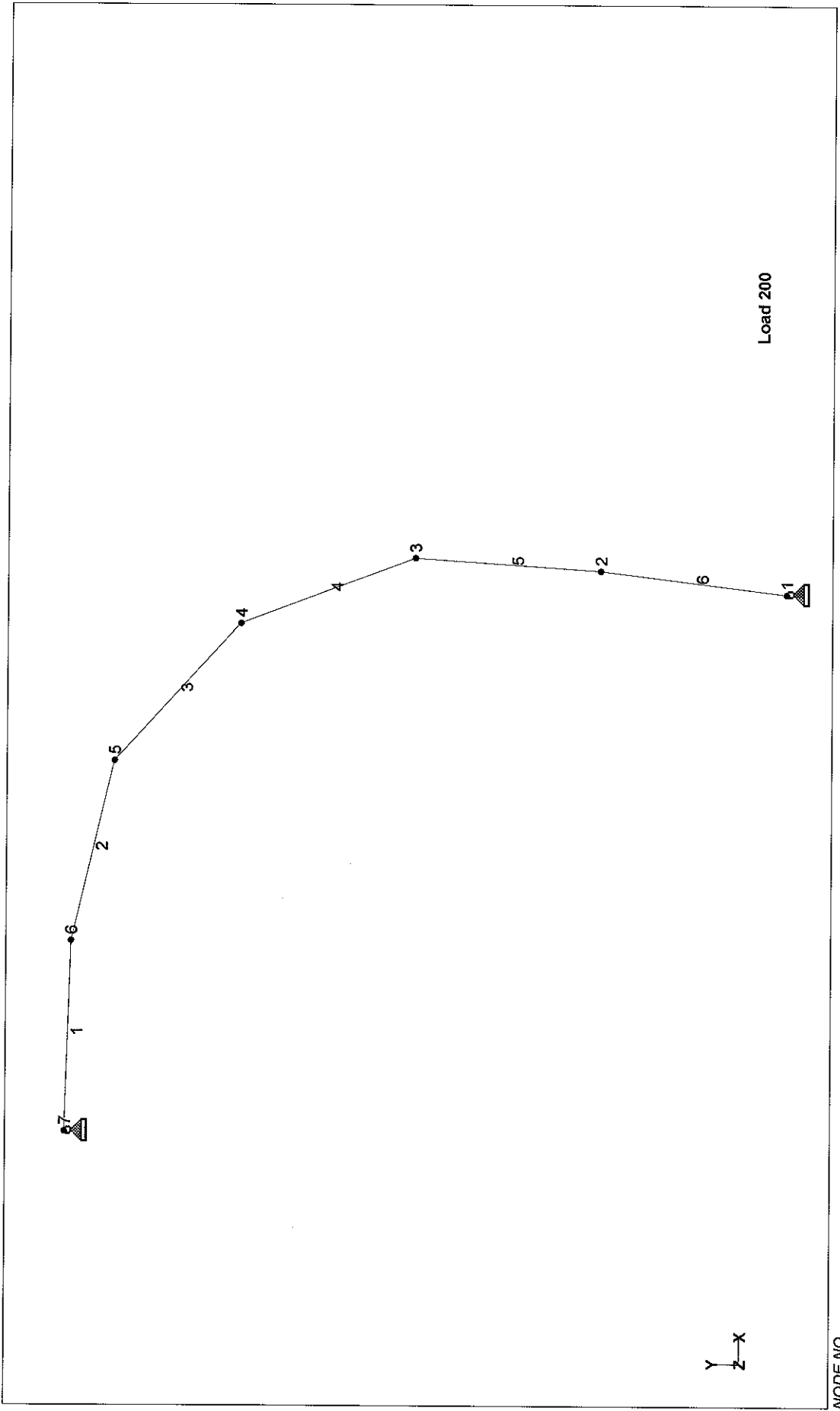
Deflection in Y	Span/360	4680/360	13
Deflection in X	HEIGHT/325	4680/325	14

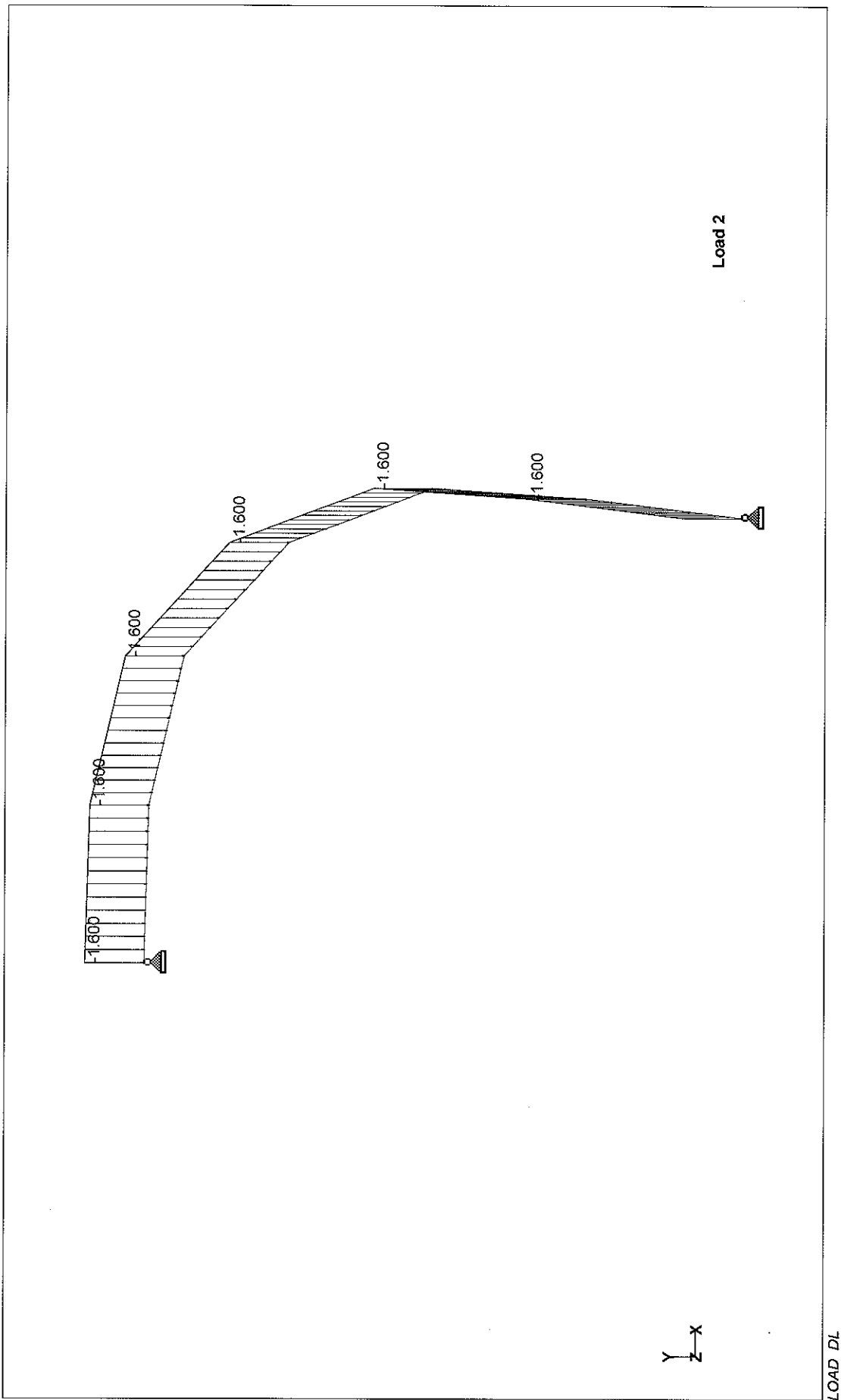
Curve Purlin IPEA120 Design :

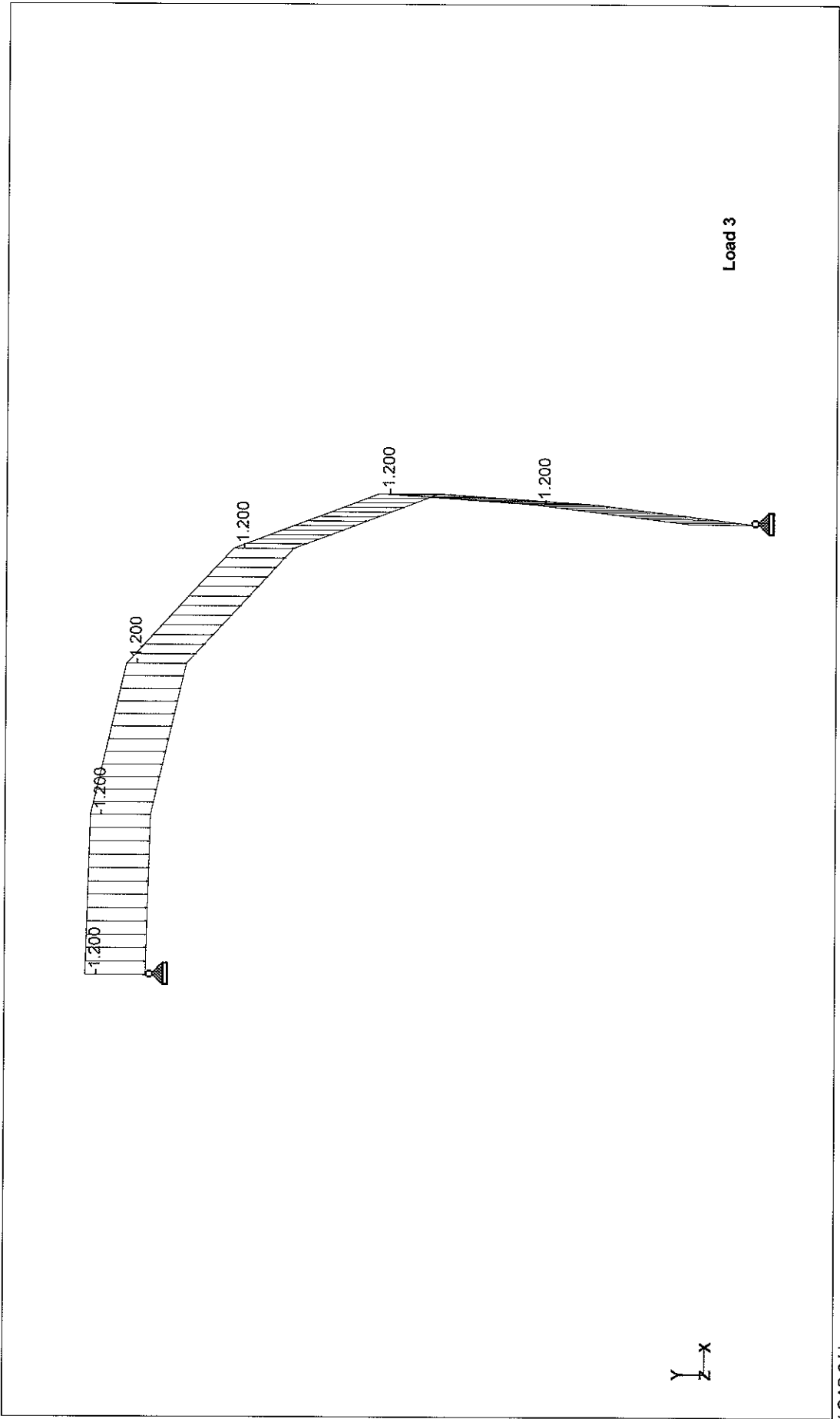
	Select a Size	=	IPEA120
	Moment of Inertia	=	257 cm ⁴
	Z xx	=	49.87 cm ³
	py	=	275 Mpa
From Staad output Data			
	Mz	=	3.6 kn-m
Moment Capacity		=	Zxx * py
		=	13.71425 kn-m
		>	3.6 kn-m

Hence O.K.

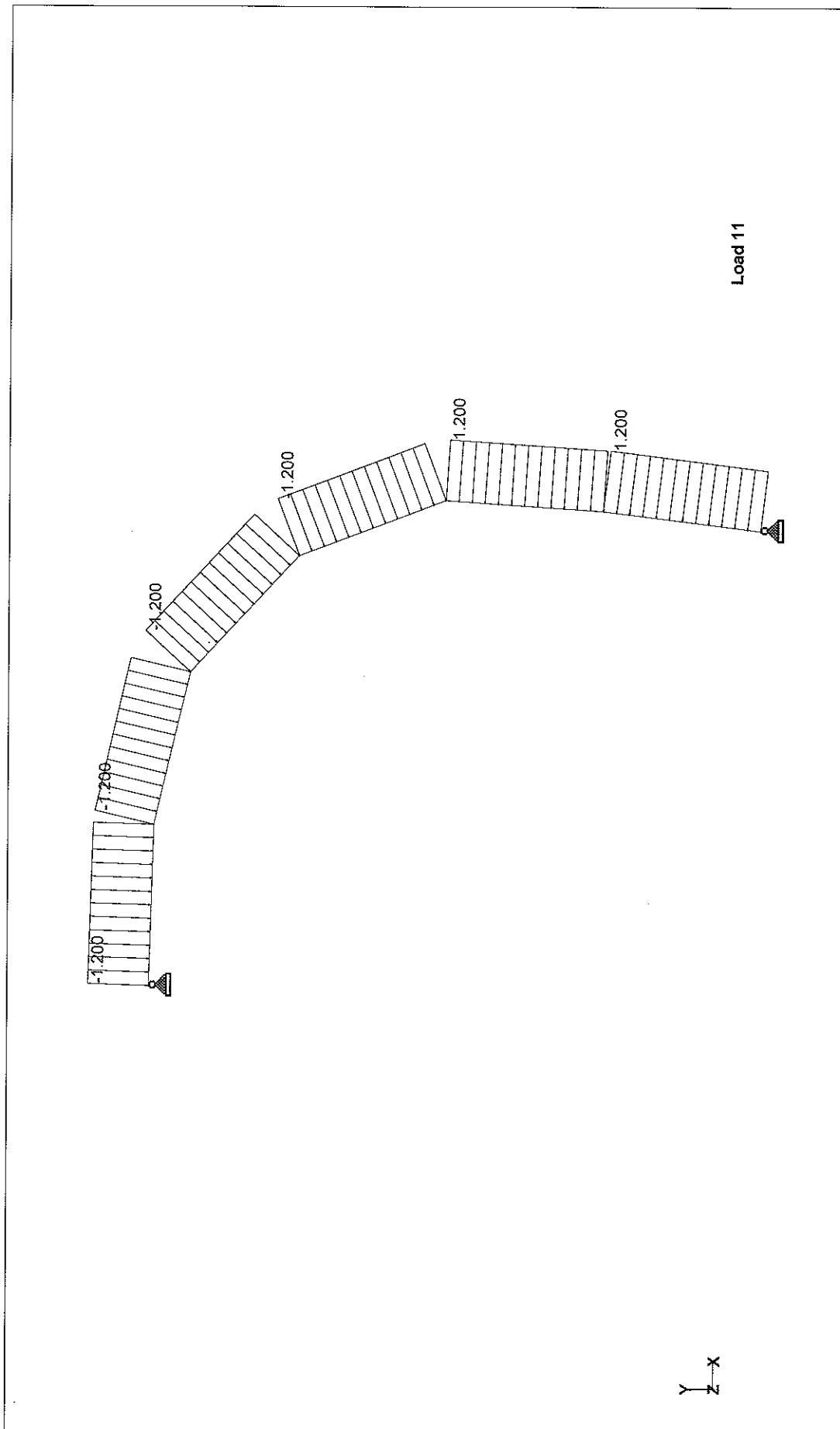




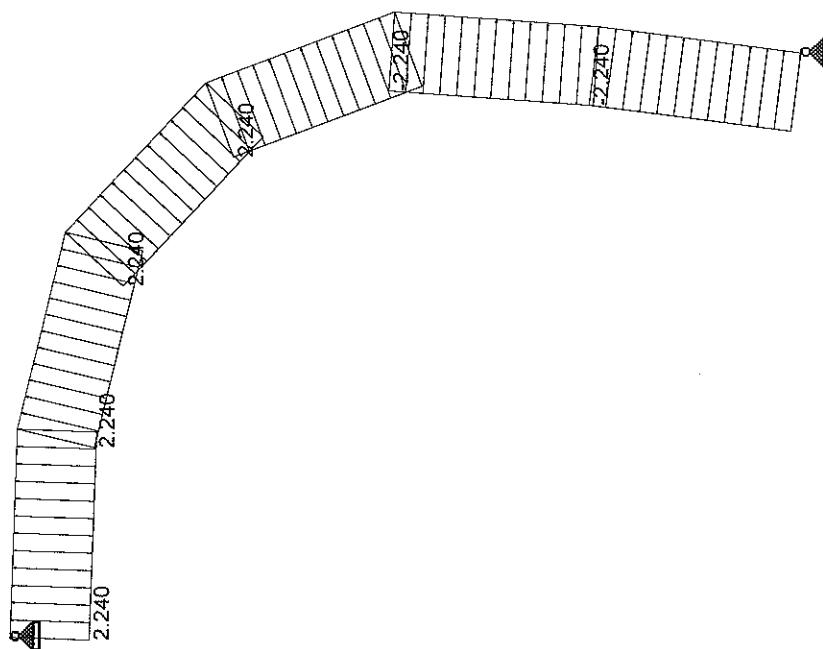




LOAD 2 LL



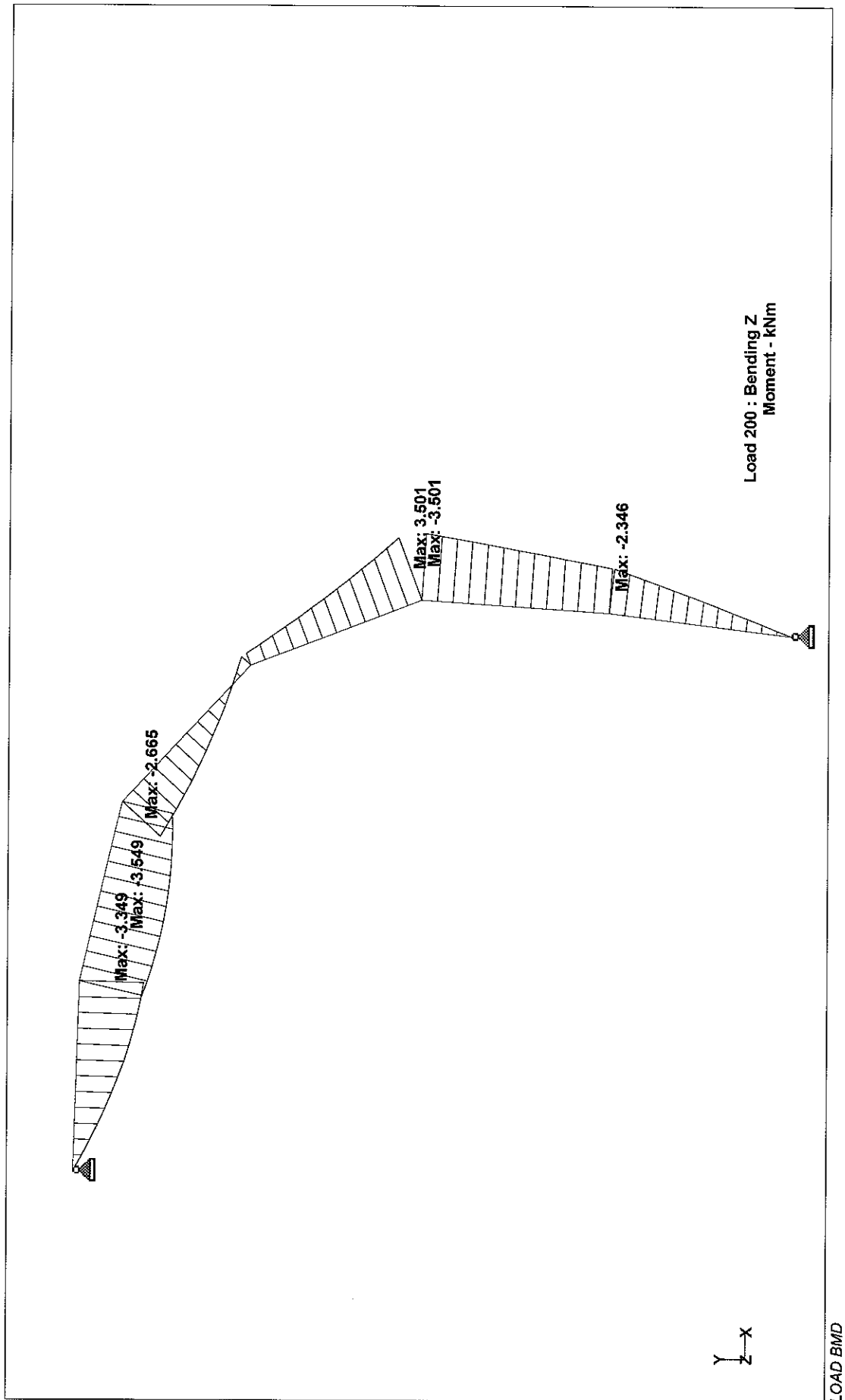
LOAD 3 WL1

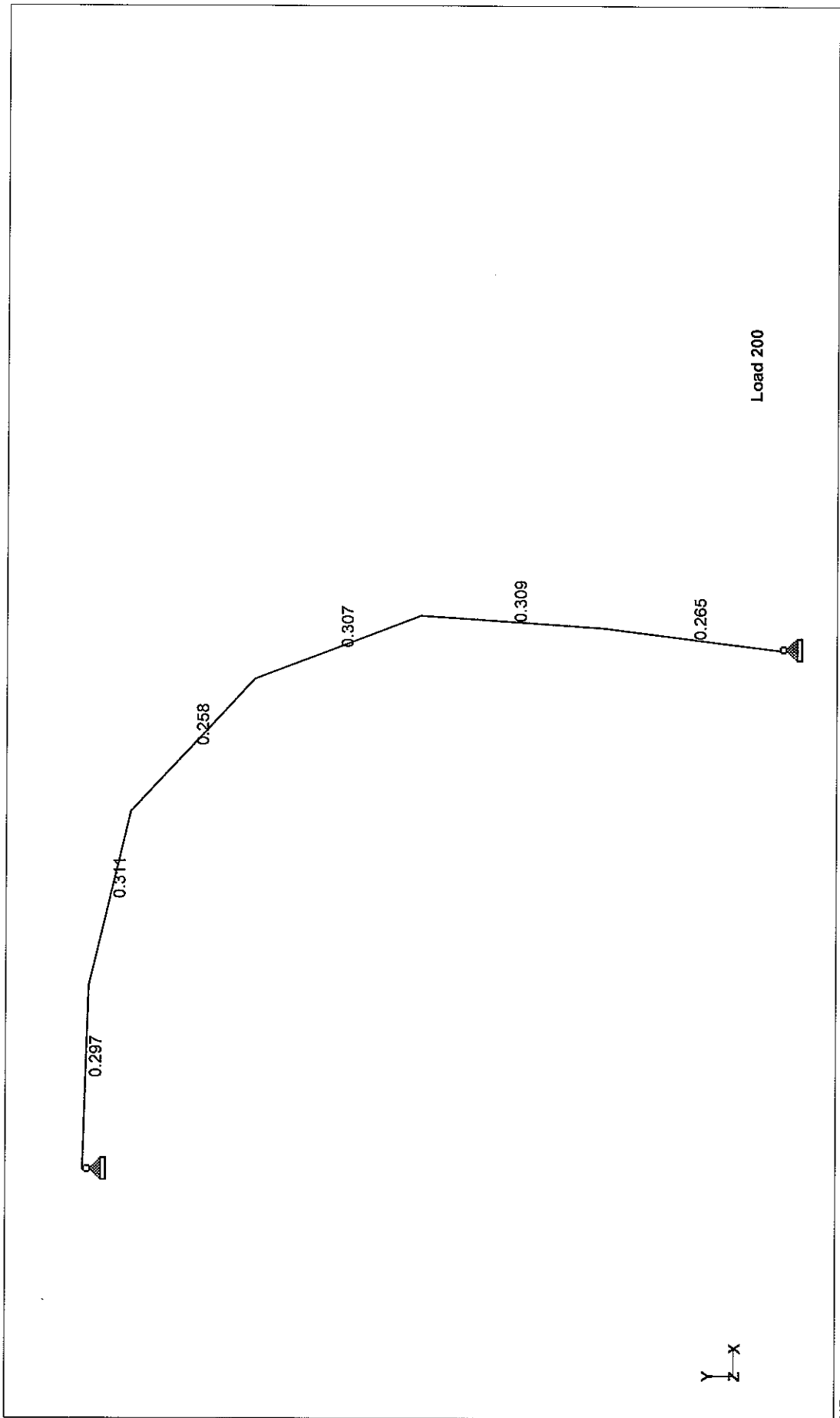


Load 12

Y
Z X

LOAD 4 WL2





□

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*****
*
*          STAAD.Pro
*          Version  2006    Bld 1002.US
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    JUN 10, 2008
*          Time=    14: 4:14
*
*          USER ID: roark
*****

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1. STAAD PLANE CURVE PURLIN
INPUT FILE: curve purlinIPEA120.STD
2. START JOB INFORMATION
3. ENGINEER DATE 20-MAY-08
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 0 0 0; 2 0.118 0.965 0; 3 0.183 1.94 0; 4 -0.159 2.84 0; 5 -0.874 3.5 0
9. 6 -1.81 3.72 0; 7 -2.8 3.75 0
10. MEMBER INCIDENCES
11. 1 7 6; 2 6 5; 3 5 4; 4 4 3; 5 3 2; 6 2 1
12. DEFINE MATERIAL START
13. ISOTROPIC STEEL
14. E 2.05E+008
15. POISSON 0.3
16. DENSITY 76.8195
17. ALPHA 1.2E-005
18. DAMP 0.03
19. END DEFINE MATERIAL
20. MEMBER PROPERTY EUROPEAN
21. 1 TO 6 TABLE ST IPEA120
22. CONSTANTS
23. MATERIAL STEEL ALL
24. SUPPORTS
25. 1 7 PINNED
26. LOAD 1 SELFWEIGHT
27. SELFWEIGHT Y -1
28. LOAD 2 DL
29. **DL =0.8 KN/M^2
30. **PURLIN SPACING = 2.0 M
31. **DL1=0.8*2.0=1.6 KN/M
32. MEMBER LOAD
33. 1 TO 6 UNI GY -1.6
34. LOAD 3 LL
35. MEMBER LOAD
36. **LL =0.6 KN/M^2
37. **PURLIN SPACING = 2.0 M
38. **LL1=0.6*2.0=1.2 KN/M
39. 1 TO 6 UNI GY -1.2
40. LOAD 11 WL1 (DN)

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41. **WIND LOAD(MAX)=0.62 KN/M^2
42. **PURLIN SPACING = 2.0 M
43. **LL1=0.62*2.0=1.24 KN/M
44. MEMBER LOAD
45. 1 TO 4 UNI Y -1.2
46. 5 6 UNI Y 1.2
47. LOAD 12 WL2 ( UP )
48. **WIND LOAD(MAX) =1.12 KN/M^2
49. **PURLIN SPACING = 2.0 M
50. **LL1=1.12*2.0=2.24 KN/M
51. MEMBER LOAD
52. 1 TO 4 UNI Y 2.24
53. 5 6 UNI Y -2.24
54. *****STEEL DESIGN *****
55. LOAD COMB 200 1.4DL+1.6LL
56. 1 1.4 2 1.4 3 1.6
57. *****WL1*****
58. LOAD COMB 201 1.2DL+1.2LL+1.2WL1
59. 1 1.2 2 1.2 3 1.2 11 1.2
60. LOAD COMB 202 1.4DL+1.4WL1
61. 1 1.4 2 1.4 11 1.4
62. LOAD COMB 203 1DL+1.4WL1
63. 1 1.0 2 1.0 11 1.4
64. *****WL1*****
65. LOAD COMB 204 1.2DL+1.2LL+1.2WL2
66. 1 1.2 2 1.2 3 1.2 12 1.2
67. LOAD COMB 205 1.4DL+1.4WL2
68. 1 1.4 2 1.4 12 1.4
69. LOAD COMB 206 1DL+1.4WL2
70. 1 1.0 2 1.0 12 1.4
71. *****DEFLECTION*****
72. LOAD COMB 301 SELF+DL
73. 1 1.0 2 1.0
74. LOAD COMB 302 LL
75. 3 1.0
76. LOAD COMB 303 0.8(LL+WL1)
77. 3 0.8 11 0.8
78. LOAD COMB 304 0.8(LL+WL2)
79. 3 0.8 12 0.8
80. *****
81. PERFORM ANALYSIS

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□

P R O B L E M S T A T I S T I C S

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NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      7/      6/      2
ORIGINAL/FINAL BAND-WIDTH=      1/      1/      6 DOF
TOTAL PRIMARY LOAD CASES =      5, TOTAL DEGREES OF FREEDOM =      17
SIZE OF STIFFNESS MATRIX =      1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE =      12.0/ 15690.2 MB

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82. LOAD LIST 301 TO 304

83. PRINT JOINT DISPLACEMENTS

☐ JOINT DISPLACE ☐☐

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = PLANE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1	301	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0024
	302	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0017
	303	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0007
	304	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0027
2	301	0.2040	-0.0279	0.0000	0.0000	0.0000	-0.0015
	302	0.1453	-0.0199	0.0000	0.0000	0.0000	-0.0011
	303	0.0665	-0.0108	0.0000	0.0000	0.0000	-0.0007
	304	0.2092	-0.0254	0.0000	0.0000	0.0000	-0.0012
3	301	0.2541	-0.0335	0.0000	0.0000	0.0000	0.0006
	302	0.1810	-0.0239	0.0000	0.0000	0.0000	0.0005
	303	0.1092	-0.0160	0.0000	0.0000	0.0000	0.0000
	304	0.2113	-0.0249	0.0000	0.0000	0.0000	0.0010
4	301	0.1147	-0.0880	0.0000	0.0000	0.0000	0.0021
	302	0.0817	-0.0627	0.0000	0.0000	0.0000	0.0015
	303	0.0611	-0.0363	0.0000	0.0000	0.0000	0.0009
	304	0.0734	-0.0761	0.0000	0.0000	0.0000	0.0016
5	301	-0.0067	-0.2204	0.0000	0.0000	0.0000	0.0012
	302	-0.0048	-0.1570	0.0000	0.0000	0.0000	0.0008
	303	-0.0018	-0.1065	0.0000	0.0000	0.0000	0.0008
	304	-0.0075	-0.1612	0.0000	0.0000	0.0000	0.0005
6	301	-0.0067	-0.2210	0.0000	0.0000	0.0000	-0.0012
	302	-0.0048	-0.1574	0.0000	0.0000	0.0000	-0.0009
	303	-0.0049	-0.1242	0.0000	0.0000	0.0000	-0.0006
	304	-0.0019	-0.1292	0.0000	0.0000	0.0000	-0.0010
7	301	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0027
	302	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0019
	303	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0016
	304	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0014

***** END OF LATEST ANALYSIS RESULT *****

84. LOAD LIST 200 TO 206

85. PRINT SUPPORT REACTION

☐ SUPPORT REACTION ☐

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
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1	200	-0.30	19.48	0.00	0.00	0.00	0.00
	201	1.49	19.07	0.00	0.00	0.00	0.00
	202	1.85	14.60	0.00	0.00	0.00	0.00
	203	1.90	11.54	0.00	0.00	0.00	0.00
	204	-3.48	9.58	0.00	0.00	0.00	0.00
	205	-3.94	3.52	0.00	0.00	0.00	0.00
7	206	-3.89	0.46	0.00	0.00	0.00	0.00
	200	0.30	5.49	0.00	0.00	0.00	0.00
	201	3.91	5.16	0.00	0.00	0.00	0.00
	202	4.45	3.87	0.00	0.00	0.00	0.00
	203	4.40	3.00	0.00	0.00	0.00	0.00
	204	-6.60	3.10	0.00	0.00	0.00	0.00
	205	-7.82	1.46	0.00	0.00	0.00	0.00
	206	-7.87	0.60	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

86. PARAMETER 1

87. CODE BS5950

88. LZ 5.4 ALL

89. PY 275000 ALL

90. TRACK 2 ALL

91. CHECK CODE ALL

□STEEL DESIGN □

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

□

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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1 ST	IPEA120	PASS	BS-4.3.6	0.297	200
		0.14 C	0.00	3.35	0.00

=====

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 274 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 99.05
 Gross Area = 11.00 Net Area = 11.00 Eff. Area = 11.00

	z-z axis	y-y axis
Moment of inertia	257.000	22.000
Plastic modulus	50.000	11.000
Elastic modulus	43.707	6.875
Effective modulus	43.707	6.875
Shear Area	5.875	4.469

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SEMI-COMPACT
 Squash Load : 302.50
 Axial force/Squash load : 0.000

	z-z axis	y-y axis
Compression Capacity	146.6	222.0
Tension Capacity	302.5	302.5
Moment Capacity	13.8	2.8
Reduced Moment Capacity	13.8	3.0
Shear Capacity	96.9	73.7

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	x-x axis	y-y axis
Slenderness	111.718	70.036
Radius of gyration (cm)	4.834	1.414
Effective Length	5.400	0.990

LTB Moment Capacity (kNm) and LTB Length (m): 11.26, 0.990

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
 Mlt = 3.35 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.075	200	-	5.5	-	-	-

□

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

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2 ST	IPEA120	PASS	BS-4.3.6	0.311	200
		0.01 C	0.00	3.55	0.00

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 274 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 96.15
 Gross Area = 11.00 Net Area = 11.00 Eff. Area = 11.00

	z-z axis	y-y axis
Moment of inertia	257.000	22.000
Plastic modulus	50.000	11.000
Elastic modulus	43.707	6.875
Effective modulus	43.707	6.875
Shear Area	5.875	4.469

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC
 Squash Load : 302.50
 Axial force/Squash load : 0.000

	z-z axis	y-y axis
Compression Capacity	146.6	226.4
Tension Capacity	302.5	302.5
Moment Capacity	13.8	2.8
Reduced Moment Capacity	13.8	2.8
Shear Capacity	96.9	73.7

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	x-x axis	y-y axis
Slenderness	111.718	67.989
Radius of gyration (cm)	4.834	1.414
Effective Length	5.400	0.962

LTB Moment Capacity (kNm) and LTB Length (m): 11.43, 0.962

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 1.00 : my = 1.00 : myx = 1.00
 Mlt = 3.55 : Mx = 3.55 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.047	201	-	3.4	-	-	-

□

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

□

3 ST	IPEA120	PASS	BS-4.3.6	0.258	204
		5.39 T	0.00	2.93	0.00

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 274 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 97.30
 Gross Area = 11.00 Net Area = 11.00 Eff. Area = 11.00

	z-z axis	y-y axis
Moment of inertia	257.000	22.000
Plastic modulus	50.000	11.000
Elastic modulus	43.707	6.875
Effective modulus	43.707	6.875
Shear Area	5.875	4.469

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC
 Squash Load : 302.50
 Axial force/Squash load : 0.018

	z-z axis	y-y axis
Compression Capacity	146.6	224.7
Tension Capacity	302.5	302.5
Moment Capacity	13.8	2.8
Reduced Moment Capacity	13.7	2.8
Shear Capacity	96.9	73.7

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	x-x axis	y-y axis
Slenderness	111.718	68.805
Radius of gyration (cm)	4.834	1.414
Effective Length	5.400	0.973

LTB Moment Capacity (kNm) and LTB Length (m): 11.36, 0.973

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 1.00 : my = 1.00 : myx = 1.00
 Mlt = 2.93 : Mx = 2.93 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.067	200	-	5.0	-	-	-

□

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
4 ST	IPEA120	PASS	BS-4.3.6	0.307	200
		6.67 C	0.00	3.50	0.00
=====					

□

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 274 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 96.28
 Gross Area = 11.00 Net Area = 11.00 Eff. Area = 11.00

	z-z axis	y-y axis
Moment of inertia :	257.000	22.000
Plastic modulus :	50.000	11.000
Elastic modulus :	43.707	6.875
Effective modulus :	43.707	6.875
Shear Area :	5.875	4.469

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC
 Squash Load : 302.50
 Axial force/Squash load : 0.022

	z-z axis	y-y axis
Compression Capacity :	146.6	226.2
Tension Capacity :	302.5	302.5
Moment Capacity :	13.8	2.8
Reduced Moment Capacity :	13.7	2.8
Shear Capacity :	96.9	73.7

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	x-x axis	y-y axis
Slenderness :	111.718	68.080
Radius of gyration (cm) :	4.834	1.414
Effective Length :	5.400	0.963

LTB Moment Capacity (kNm) and LTB Length (m): 11.42, 0.963

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 1.00 : my = 1.00 : myx = 1.00
 Mlt = 3.50 : Mx = 3.50 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.052	204	-	3.9	-	-	-

□

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

□

5 ST	IPEA120	PASS	BS-4.3.6	0.309	200
		11.10 C	0.00	3.50	0.00

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 274 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 97.72
 Gross Area = 11.00 Net Area = 11.00 Eff. Area = 11.00

	z-z axis	y-y axis
Moment of inertia	257.000	22.000
Plastic modulus	50.000	11.000
Elastic modulus	43.707	6.875
Effective modulus	43.707	6.875
Shear Area	5.875	4.469

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC
 Squash Load : 302.50
 Axial force/Squash load : 0.037

	z-z axis	y-y axis
Compression Capacity	146.6	224.1
Tension Capacity	302.5	302.5
Moment Capacity	13.8	2.8
Reduced Moment Capacity	13.7	2.8
Shear Capacity	96.9	73.7

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	x-x axis	y-y axis
Slenderness	111.718	69.096
Radius of gyration (cm)	4.834	1.414
Effective Length	5.400	0.977

LTB Moment Capacity (kNm) and LTB Length (m): 11.34, 0.977

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 1.00 : my = 1.00 : myx = 1.00
 Mlt = 3.50 : Mx = 3.50 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.033	206	-	2.4	-	-	-

□

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
6 ST	IPEA120	PASS 5.74 C	BS-4.3.6 0.00	0.265 3.02	204 0.00

□

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 274 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 97.22
 Gross Area = 11.00 Net Area = 11.00 Eff. Area = 11.00

	z-z axis	y-y axis
Moment of inertia	257.000	22.000
Plastic modulus	50.000	11.000
Elastic modulus	43.707	6.875
Effective modulus	43.707	6.875
Shear Area	5.875	4.469

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC
 Squash Load : 302.50
 Axial force/Squash load : 0.019

	z-z axis	y-y axis
Compression Capacity	146.6	224.8
Tension Capacity	302.5	302.5
Moment Capacity	13.8	2.8
Reduced Moment Capacity	13.7	2.8
Shear Capacity	96.9	73.7

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	x-x axis	y-y axis
Slenderness	111.718	68.744
Radius of gyration (cm)	4.834	1.414
Effective Length	5.400	0.972

LTB Moment Capacity (kNm) and LTB Length (m): 11.37, 0.972

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 1.00 : my = 1.00 : myx = 1.00
 Mlt = 3.02 : Mx = 3.02 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.063	204	-	4.6	-	-	-

***** END OF TABULATED RESULT OF DESIGN *****

92. STEEL TAKE OFF ALL

☐ STEEL TAKE OFF ☐

☐

STEEL TAKE-OFF

PROFILE	LENGTH (METE)	WEIGHT (KN)
ST IPEA120	5.84	0.493

TOTAL =		0.493

***** END OF DATA FROM INTERNAL STORAGE *****

93. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JUN 10,2008 TIME= 14: 4:14 ****

(Grid 1-1 To 1-52)

[illegible]

Max. Axial Force in Curve Purlin =	20	kN
Shear force per bolt	=	20 / 2
	=	10.00 kN

As the Force is very small, but we have to provide the nominal bolt size for stability

i.e. Provide		2 No. Of	M-12	Grade-8.8 Bolts
--------------	--	----------	------	-----------------

Shear force per bolt = 31 KN > 10.0 KN
Hence ok

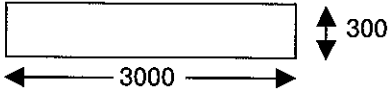
3. Buffer Stop Design

Document No.: DM001-D-GAB-WSC-DR-DCC-545046	Date: 11 Jun 2008
Rev A1	

Track Slab Design (Check for Buffer Stop)

In plane Tension & Compression force in slab due to Buffer Stop in Buffer Zone Area

a) Check for Tension

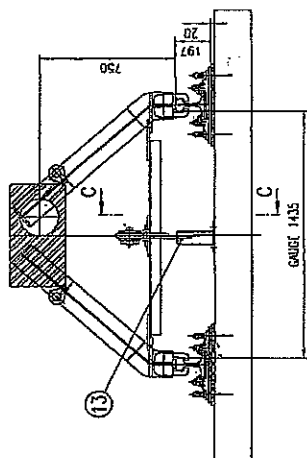
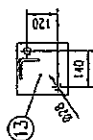
i) Force in Longitudinal Direction (IIR to Rail)	=	800 KN	
Tensile Force (T)	=	800 KN (assumed to be acting on 3m width containing track)	
Tension Capacity of R/F bars	=	$0.6 f_y \times A_{st}$	
Tension Capacity of T12 bars	=	$0.6 \times 460 \times 113$	
	=	31.2 KN	
			
Spacing of extra bars provided as per detail in Dwg.# DM001-D-GAB-WSC-DD-DCC-551040	=	T12 @ 100	T & B
No. of bars in 3000 mm width	=	31 x 2	
Total Tension Capacity of extra R/F in the effective width of slab	=	$31.2 \times 31 \times 2$	
	=	1934.4 KN	> 800 KN Hence O.K.

b) Check for Compression

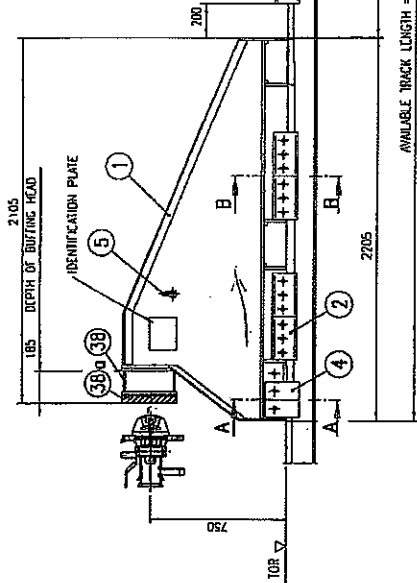
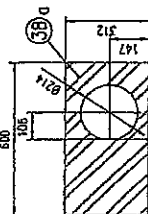
Compressive Force (F_c)	=	800	KN	
Spacing of extra bars provided as per detail in Dwg.# DM001-D-GAB-WSC-DD-DCC-551040	=	T12 @ 100	T & B	
No. of bars in 3000 mm width	=	31 x 2		
Steel Area (A_{sc})	=	$31 \times 2 \times 113$		
	=	7006.00	mm ²	
X-section Area of Concrete (A_c)	=	$3000 \times 300 - 7006$		
	=	892994.00	mm ²	
	=	$0.35 f_{cu} A_c + 0.7 A_{sc} f_y$		
Total Ultimate Compression Capacity (P_u)	=	$0.35 \times 40 \times 892994 + 0.7 \times 7006 \times 460$		
	=	14757	KN	
Total Compression Capacity (P)	=	9838.00	KN	> 800 Hence O.K.

ALL UNITS ARE IN MM, UNLESS OTHERWISE SPECIFIED.

STOPPER
SCALE 1:10



BUFFING HEAD
SCALE 1:30

TYPE: UIC 54
DURATION: 1:20

STOPPING DISTANCE

UNAVAILABLE TRACK LENGTH = 7400

IDENTIFICATION PLATE

SCALE 1:1

IDENTIFICATION PLATE
IN ACCORDANCE TO
DWS. NO. 7240\7-21

THE DATA ON THIS LABEL IS VALID ONLY FOR THE ORIGINAL SPECIFICATION. REQUEST WHICH THIS EQUIPMENT WAS SUPPLIED. IN THE EVENT OF THE EQUIPMENT BEING REMOVED FROM ITS ORIGINAL COMMISSIONING POINT NAME SHALL IN NO WAY BE LIABLE FOR THE CONSEQUENCES RESULTING FROM SUCH A REMOVAL.

SIGNAL NO./TYPE

EC-0000	6/7/92 For Depot Use
194 orders no.: DMV-88-76-101-200	
TP=70.25 / 135.11 / 167.26; max. v=135.11	
Sig. distance from end of Station Stop to last track end = 516m	

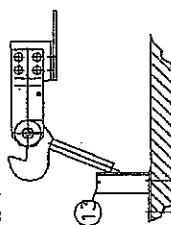
5/24/10 / Year 2007

11:59 PM CST

MANUFACTURER:
AARAS GmbH & Co. KG
P.O. Box 35 28, 49075 Osnabrück
Germany
Tele: +49 (0)5 41 / 91 20 7 - 0
Fax: +49 (0)5 41 / 91 20 7 - 111

SECTION C-C

SCALE 1:10

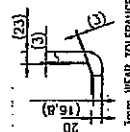


91 JANUARY 16 1966

- BLAST CLEANED
- GALVANIZED (WEL. 60 Am)
- SHEET WITH NON-FIBROUS SHEET
- 1 PRIMER COAT (WEL. 30 Am)
- 1 INTERMEDIATE COAT (WEL. 60 Am)
- 1 COATED TOP COAT (WEL. 60 Am)

DETAIL X

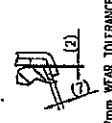
SCALE 1-5



Wear Tolerance

DETAIL Y

SCALE 1:5



10-YEAR TOLERANCE

[illegible]

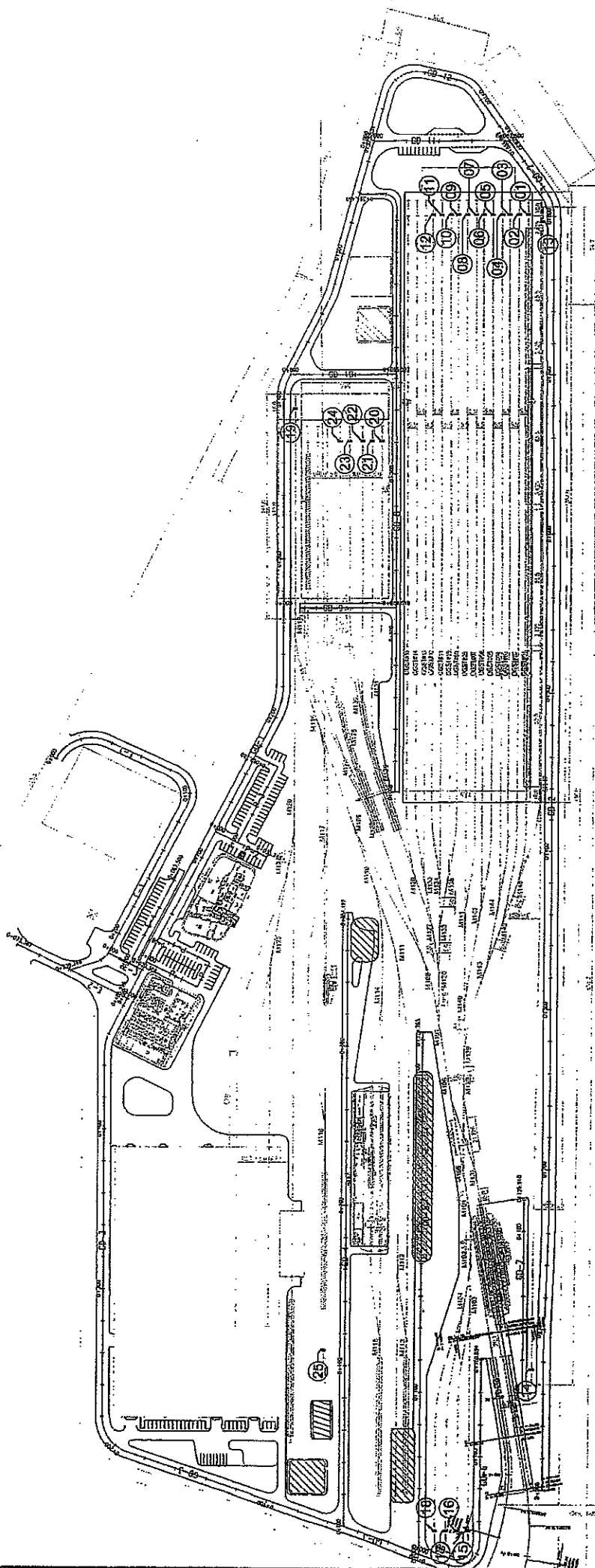
.) STARTING TRACK DASHED LINE ARE TO BE CONSTRUCTED DURING PHASE 2, ON FUTURE (B) ACQUISITIONS.

2. AFTER STOP.

TRUCK FUNCTION	ENTER STOP
DIS #1	1
STORAGE TRUCK #1	2
DIS #2	3
STORAGE TRUCK #2	4
DIS #3	5
STORAGE TRUCK #3	6
DIS #4	7
STORAGE TRUCK #4	8
DIS #5	9
STORAGE TRUCK #5	10
DIS #6	11
STORAGE TRUCK #6	12
DIS #7	13
STORAGE TRUCK #7	14
DIS #8	15
STORAGE TRUCK #8	16
DIS #9	17
STORAGE TRUCK #9	18
DIS #10	19
STORAGE TRUCK #10	20
DIS #11	21
STORAGE TRUCK #11	22
ST #1	23
ST #2	24
ST	25
DIS	26
DIS	27
DIS	28
DIS	29
DIS	30
DIS	31
DIS	32
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DIS	99
DIS	100

ENTER STOP TOTAL INQUIRIES - 25 575

PUBLIC AREA
(86,818.579 M²)



LOGO:  CONSORTIUM: Dubai Rapid Link Consortium A JV of RAIL INVESTMENT CORPORATION and AL FALAKH INVESTMENT CO. LLC A. International Corporation										STATUS: PRELIMINARY										PROJECT: PROJECT DM-001 DUBAI METRO PROJECT																			
LOC:  DUBAI, U.A.E.										DATE: 15/01/2008										DESIGNED BY: R. INABAHA										ISSUING TITLE: GREEN LINE AL QUSAYS DEPOT BUFFER STOP LAYOUT									
TOGETHER:  RAIL INVESTMENT CORPORATION AL FALAKH INVESTMENT CO. LLC										SCALE: 1:1										CHECKED BY: M. KUSUMODHIL HAZAM										DATE: 11/01/2008									
REVISIONS: NO. DATE DESCRIPTION										APPROVED BY: [Signature]										DATE: 11/01/2008										ISSUING NO: DM001-E-ALL-RSW-00-DCR-668112									



DUBAL METRO PROJECT

MATERIAL SPECIFICATION

FOR

BUFFER STOPS

Ref.



MATERIAL SPECIFICATION FOR BUFFER STOP**TABLE OF CONTENTS**

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ATTACHMENT-1; PASSENGER CAPACITY INFORMATION



1 GENERAL

1.1 Design Condition

General design condition in this project is as follows.

Description	Value	Description	Value
Gauge	1435 mm	Rail Profile & Inclination	UIC54E1 & 1/20 or Vertical
Annual Tonnage	45x10 ⁶ ton/year	Climate Condition Air Temperature	Max +52°C
Design Speed	100 kph		Min +1°C
Train Max Operation Speed (Mainline)	90 kph	Climate Condition Ambient Humidity	Max 100%
Train Max Operation Speed (Depot Connection Line)	40 kph		Min 20%
Train Max Operation Speed (Depot)	25 kph	Train Power Supply	750 volt DC
Train Max Axle Load	14 tonnes	Gauge Widening (Slak)	190m ≤ R < 300m
Minimum Curve Mainline	250m		140m ≤ R < 190m
Minimum Curve Depot	140m		

Vehicle Coupler Allowable Collision Force; 800kN

Collision Speed for Mainline; 18km/h, for Depot; 18km/h (*1)

(*1) Note; Collision condition (speed and mass) will be coordinated with operation / signalling / rolling stock.